

The SymC Neurostability Principle:

Governing Physics of Instability Architectures in Alzheimer’s and Dementia

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Abstract

Dementia subtypes share terminal phenotypes but diverge in their dynamical origins. This paper presents a unified Symmetrical Convergence (SymC) framework that reframes all major dementias as distinct failures of viscoelastic control, measurable through the Adaptive Susceptibility Ratio $\chi = \gamma/(2|\omega|)$. The framework identifies four primary failure types: **Type 1 (Alzheimer’s disease)** involves stiffness collapse producing a U-shaped $\chi(t)$ trajectory; **Type 2 (Lewy Body Dementia)** involves controller failure with preserved stiffness but pathological damping variance; **Type 3 (Vascular Dementia)** involves impedance mismatch at vessel-tissue interfaces; and **Type 4 (Frontotemporal Dementia)** involves topology failure through network decoupling. By introducing χ as a universal translation layer, the framework reveals that the damping ratio in control theory (ζ), the viscoelastic loss factor in materials science (ξ), and the stability fluctuations observed clinically are all measuring the same quantity. The Viscoelastic Cascading Failure Model (VCFM) provides the mechanism for Type 1: neural tissue acts as a low-pass filter, with stiffness decline producing a predictable spectral cascade (gamma scattering $\rightarrow$ alpha detuning $\rightarrow$ theta dominance). Subtype-specific instability indices precede clinical diagnosis by 3 to 15 years, validated by MR elastography, EEG variability metrics, and clinical fluctuation data. This unified framework enables physics-based differential diagnosis in the preclinical window, when protein pathology remains ambiguous, and establishes dementia not as “loss of brain” but as “loss of stability”: a problem amenable to detection, measurement, and potentially, correction.

1 Introduction

1.1 The Silo Problem

A fundamental barrier to progress in neurodegeneration research is disciplinary fragmentation. The same stability principle appears across fields under different names: control theory measures the **damping ratio** ζ ; materials science measures the **viscoelastic loss factor** ξ ; information theory measures entropy production and signal-to-noise ratios; clinical neurology measures cognitive fluctuation and variability indices.

A physicist sees ζ , a biologist sees ξ , a clinician sees “good days and bad days,” and they do not recognize they are observing the same phenomenon. This paper introduces χ as the **universal translation layer**: a meta-parameter that maps to the specific observables of each domain while revealing their mathematical equivalence.

1.2 The Diagnostic Problem

Current diagnostic frameworks for dementia rely on static thresholds (amyloid PET positivity, CSF p-tau elevation, cognitive test cutoffs) that identify the consequences of disease after significant, often irreversible damage has occurred. The “one gene, one protein, one disease” model that has dominated neurology fails to account for the dynamical instability that frequently precedes static decline: patients in preclinical

stages report fluctuations in memory and attention, gait variability, sleep fragmentation, and “good days and bad days.”

Yet clinical reality is dominated by mixed pathologies and overlapping phenotypes. **This instability is not merely a symptom but the primary measurable pathology.** The Symmetrical Convergence (SymC) framework posits that stable neural function requires the brain to maintain dynamics within an Adaptive Window ($\chi \approx 0.8$ to 1.0), where the system balances dissipation against restoration.

Moreover, emerging multi-modal data reveals that the “AD Cascade” is not universal. Other dementias do not follow the same path because they do not suffer the same physical failure:

- **DLB Paradox:** Patients show massive cognitive fluctuations but normal brain stiffness on MR elastography.
- **FTD Paradox:** Patients show severe behavioral collapse but normal posterior EEG power spectra for years.
- **VaD Pattern:** Stepwise decline follows structural events, not intrinsic oscillatory instability.

These observations demand a unified framework capable of distinguishing failure modes *before* clinical phenotypes converge into indistinguishable “dementia.”

2 Theoretical Foundation: The Universal Adaptive Parameter

2.1 The Neurostability Principle

The Neurostability Principle states that neural systems remain functional only while their domain-specific adaptive susceptibility ratios (χ_i) remain within the Adaptive Window ($\chi \approx 0.8$ – 1.0); dementia arises when stiffness, damping, impedance, or network topology drive χ outside this stability boundary, producing subtype-specific instability architectures.

The remainder of this section derives this principle mathematically and establishes its cross-domain validity.

2.2 Definition of χ

The dimensionless parameter χ (Chi), defined as the **Adaptive Susceptibility Ratio**, represents the universal ratio of dissipation (damping) to restoration (stiffness) in any adaptive system:

$$\chi(t) = \frac{\gamma(t)}{2|\omega(t)|} \quad (1)$$

where γ represents effective damping (inhibition, friction, dissipation) and ω represents characteristic frequency (drive, excitation, neural oscillation).

The mathematical realization adapts to each domain while the stability principle remains invariant:

- **Dynamical Control (EEG):** χ maps to the damping ratio $\zeta = \gamma/(2\omega)$
- **Structural Physics (MRE):** χ maps to the viscoelastic damping ratio $\xi = G''/(2G')$
- **Information Theory:** χ acts as an inverse proxy for signal-to-noise ratio

2.3 The Adaptive Window

The value of χ determines system behavior:

- $\chi < 0.8$: **Underdamped:** oscillatory, unstable, “noisy”
- $\chi \approx 0.8$ to 1.0 : **Adaptive window:** optimal efficiency
- $\chi > 1.2$: **Overdamped:** rigid, slow, “stuck”

The optimal operating point is $\chi \approx 0.9$ rather than exactly 1.0 . This slight underdamping provides “observability”; the system retains enough oscillatory behavior to detect and adapt to environmental changes.

2.4 Information-Theoretic Foundation

The adaptive window is not arbitrary. The Adaptive Intelligence Framework (AIF) demonstrates that $\chi = 1$ represents the **information-theoretic optimum** where systems maximize information gain while minimizing entropy production [18]. At this boundary, the discriminant $\Delta = \gamma^2 - 4\omega^2$ vanishes, producing the fastest non-oscillatory response. Derivation from the Lindblad master equation shows that measurement-induced decoherence and environmental dissipation achieve optimal balance precisely at $\chi = 1$ (see Supplementary Materials S6).

Under realistic constraints (feedback delays, finite bandwidth, measurement noise), the operational optimum shifts slightly below the boundary to $\chi \approx 0.8\text{--}0.95$. This explains why healthy neural systems cluster in this range: they have evolved to maximize information processing efficiency under biological constraints. Dementia, in this framework, represents not merely “instability” but **loss of information efficiency**: the brain’s capacity to extract signal from noise degrades as χ deviates from the adaptive window.

2.5 Multi-Domain Extension

For complex disorders affecting multiple brain systems, a single scalar χ is insufficient. Domain-specific ratios are defined as:

$$\chi_i(t) = \frac{\gamma_i(t)}{2|\omega_i(t)|}, \quad i \in \{\text{cog, motor, exec, ...}\} \quad (2)$$

The overall cognitive status may follow a weakest-link model:

$$\chi_{\text{eff}} = \text{argmin}_i |\chi_i - 1|^{-1} \quad (3)$$

such that failure in *any* domain crossing threshold triggers clinical impairment.

2.6 Cross-Domain Convergence

Independent measurements converge on the same physical boundaries:

Table 1: Cross-domain convergence on the χ boundary.

Domain	Variable	Equation	χ Equivalence
Control Theory	Damping ratio ζ	$\zeta = \gamma/(2\omega)$	$\chi \equiv \zeta$
Materials Science	Loss factor ξ	$\xi = G''/(2G')$	$\chi \equiv \xi$
Clinical	Variability index	Fluctuation metrics	χ deviation

This convergence resolves the silo problem. A physicist measuring ζ , a materials scientist measuring ξ , and a clinician observing “good days and bad days” are all detecting the same phenomenon: deviation from the $\chi \approx 1$ boundary.

3 Failure Mode Taxonomy

Four primary failure modes exist, each corresponding to disruption of a specific variable and governed by distinct dynamical equations:

3.1 Stiffness Collapse (SC)

The restoring force degrades:

$$\omega_{\text{eff}}^2(t) = \frac{G'(t)}{\rho} \rightarrow 0 \quad \text{as} \quad G'(t) \downarrow \quad (4)$$

Tissue softening (AD, FTD) reduces the effective spring constant, slowing oscillations and eventually eliminating resonant structure.

3.2 Controller Failure (CF)

The gain becomes stochastic:

$$\dot{\chi} = -\frac{1}{\tau}(\chi - \chi_{\text{set}}) + \sigma\xi(t) \quad (5)$$

where τ is the relaxation time, $\chi_{\text{set}} \approx 0.9$ is the homeostatic setpoint, and $\xi(t)$ is white noise. In DLB, the noise amplitude σ exceeds the restoring force $1/\tau$, preventing convergence to the attractor.

3.3 Impedance Mismatch (IM)

Energy reflection at interfaces:

$$R = \frac{Z_{\text{tissue}} - Z_{\text{vessel}}}{Z_{\text{tissue}} + Z_{\text{vessel}}}, \quad Z = \sqrt{\rho G'} \quad (6)$$

In health, $Z_{\text{tissue}} \approx Z_{\text{vessel}}$ (matched load, $R \approx 0$). In VaD, vessel stiffening and tissue softening drive $|R| \rightarrow 1$, causing destructive wave interference.

3.4 Topology Failure (TF)

Loss of global synchronization:

$$r(t) = \left| \frac{1}{N} \sum_{j=1}^N e^{i\theta_j(t)} \right| \rightarrow 0 \quad (7)$$

The Kuramoto order parameter r measures global phase coherence. In FTD, $r \rightarrow 0$ (decoherence) despite preserved local power spectral density.

The SymC Dementia Taxonomy <i>Differential Diagnosis via Instability Architecture</i>				
Dementia Subtype	Physics Failure Mode	SymC Variable	Primary Biomarker	Lead Time
Alzheimer's (AD)	Stiffness Collapse (Structural Failure)	Stiffness (G')	Alpha Detuning (Spectral Slowing)	3-5 Years
Lewy Body (DLB)	Controller Failure (Gain Instability)	Damping (y)	Freq. Variability (DFV / Fluctuations)	10-15 Years
Vascular (VaD)	Impedance Mismatch (Energy Reflection)	Impedance (ΔZ)	Focal Slowing & Gait Velocity	5+ Years
Frontotemporal (FTD)	Topology Failure (Network Decoupling)	Connectivity (C)	Silent EEG + Behavioral Change	5-8 Years

Figure 1: **Figure 1: The SymC Dementia Taxonomy.** Summary of failure modes mapped to their governing variables, primary biomarkers, and preclinical lead times. Each subtype fails through a distinct physical mechanism, enabling differential diagnosis via targeted measurement.

4 Type 1 Failure: Alzheimer's Disease (The Stiffness Cascade)

AD serves as the reference case for the SymC framework because it represents the most thoroughly characterized dementia and exhibits all key features of stiffness collapse.

4.1 The Two-Phase Failure Model

AD progression follows a two-phase dynamical failure:

Phase 1, Underdamped Instability ($\chi < 0.8$): Breakdown of inhibitory regulation (GABAergic loss) leads to hyper-responsive, noisy dynamics. Clinically, this manifests as cognitive fluctuations, anxiety, and sleep fragmentation (Preclinical/MCI).

Phase 2, Overdamped Rigidity ($\chi > 1.2$): Compensatory mechanisms (amyloid accumulation, glial scarring) massively increase effective damping. The system becomes sluggish and rigid, manifesting as monotonic cognitive decline and apathy (Dementia).

The resulting trajectory is **U-shaped**: $\chi(t)$ begins at ~ 0.9 (healthy), dips below 0.8 (MCI instability), then rises above 1.2 (AD rigidity). This unifies the paradox that early AD shows hyperactivity while late AD shows hypoactivity.

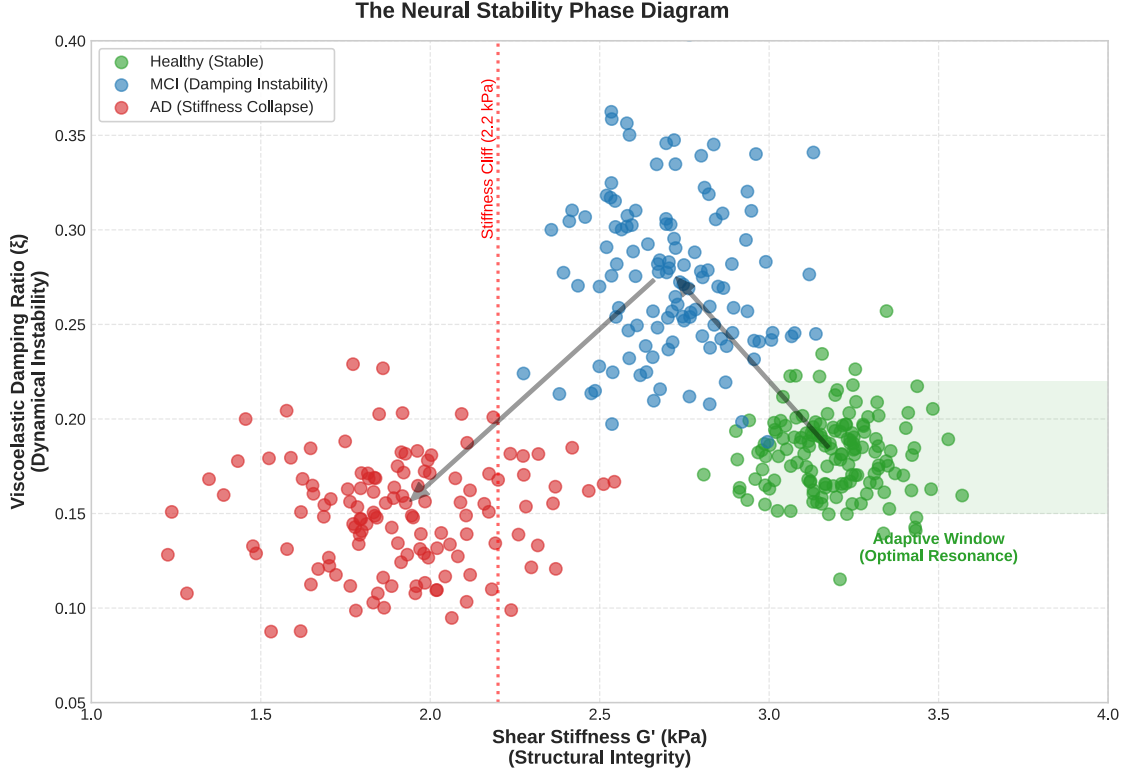


Figure 2: **Figure 2: The Neural Stability Phase Diagram.** Alzheimer’s progression as a trajectory through the Stability Plane defined by Shear Stiffness G' (Structural Integrity) and Viscoelastic Damping Ratio ξ (Dynamical Instability). Healthy brains (green) occupy the high-stiffness/optimal-damping Adaptive Window. MCI (blue) manifests as a damping spike before structural collapse, with trajectories crossing the Stiffness Cliff at 2.2 kPa. AD (red) represents terminal stiffness failure. The gray trajectory arrow shows the characteristic path: stable $\rightarrow$ damping instability $\rightarrow$ structural collapse.

4.2 MRE: The Central Equation

MRE measures shear wave propagation through brain tissue, yielding the Complex Shear Modulus $G^* = G' + iG''$, where G' (storage modulus) represents stiffness and G'' (loss modulus) represents viscosity. The **Viscoelastic Damping Ratio** is:

$$\xi = \frac{G''}{2G'} \equiv \chi \quad (8)$$

Note: This definition uses $\xi = G''/(2G')$, the viscoelastic *damping ratio*, rather than the raw loss tangent $\tan \delta = G''/G'$ sometimes used in materials science. The factor of 2 ensures direct equivalence with the control-theoretic damping ratio $\zeta = \gamma/(2\omega)$.

This equivalence is the core unification. The ratio of dissipation to restoration measured in tissue mechanics is mathematically identical to the ratio measured in neural dynamics.

4.3 The Stiffness Collapse

In AD, cytoarchitectural degradation manifests as a measurable collapse in Storage Modulus (G'). Meta-analysis of clinical MRE data establishes approximate thresholds:

- Healthy: $G' > 3.0$ kPa
- At-Risk: $G' \approx 2.5$ kPa
- AD Lesion: $G' < 2.2$ kPa

4.4 The Lead-Time Principle

A critical finding from longitudinal MRE research is that **damping changes precede stiffness changes**. The viscoelastic damping ratio ξ begins to rise while tissue stiffness G' remains normal. Only later, typically 3 to 5 years, does the structural collapse become measurable. This provides a biomarker ($\chi \equiv \xi$) that detects disease during the reversible phase.

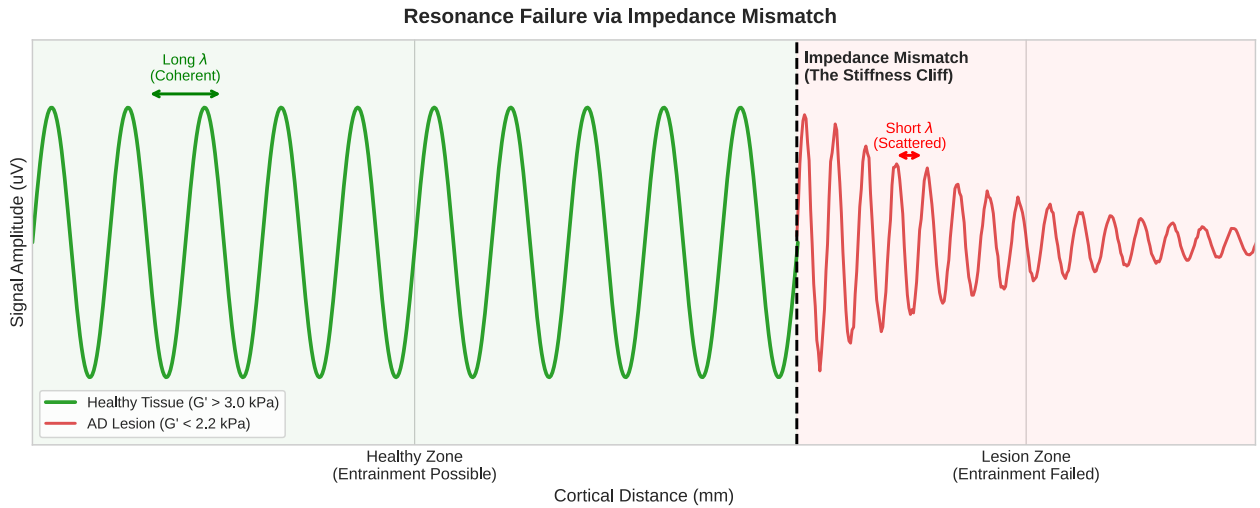


Figure 3: **Figure 3: Resonance Failure via Impedance Mismatch.** In healthy tissue ($G' > 3.0$ kPa, green), long wavelengths propagate coherently across cortical distance. At the impedance mismatch boundary (the Stiffness Cliff), signals entering the lesion zone ($G' < 2.2$ kPa, red) experience wavelength contraction and scattering. This explains why gamma entrainment therapies fail in advanced AD: the signal cannot penetrate damaged tissue.

4.5 The Viscoelastic Cascading Failure Model (VCFM)

The relationship between stiffness and wavelength ($\lambda \propto \sqrt{G'}$) implies that neural tissue acts as a **mechanical low-pass filter**. As stiffness (G') declines, the maximum supportable frequency drops. This provides a mechanistic explanation for EEG slowing in AD, unifying electrophysiology and tissue biomechanics under a single physical model.

Three critical modes diagnose the scale of mechanical failure:

Micro-Structural Scattering Mode (Gamma / 40 Hz): Due to its short wavelength, this mode is sensitive to micro-scale impedance mismatches. Loss of 40 Hz coherence indicates micro-structural softening ($G' \approx 2.8$ kPa), where local lesions act as scattering centers even before gross atrophy occurs.

Fundamental Resonant Mode (Alpha / 10 Hz): This represents the primary eigenfrequency of the cortical network. As tissue stiffness drops further ($G' \approx 2.5$ kPa), the structural resonance mechanically

detunes, forcing the peak frequency to shift downward (e.g., 10 Hz $\rightarrow$ 8 Hz). This alpha-shift is one of the earliest and most reliable EEG predictors of AD conversion; the VCFM provides the physical mechanism: the brain’s resonant cavity is literally softening.

Viscoelastic Compliance Mode (Theta / 6 Hz): As the tissue enters a high-compliance state ($G' < 2.0$ kPa), only long wavelengths can propagate without significant attenuation. The “theta dominance” seen in AD is physically explained as a low-pass filter effect: it is the residual mode that remains after higher-frequency capacity is lost.

The VCFM spectral cascade (scattering $\rightarrow$ detuning $\rightarrow$ compliance dominance) represents a predictable sequence determined by tissue mechanics, not random degeneration.

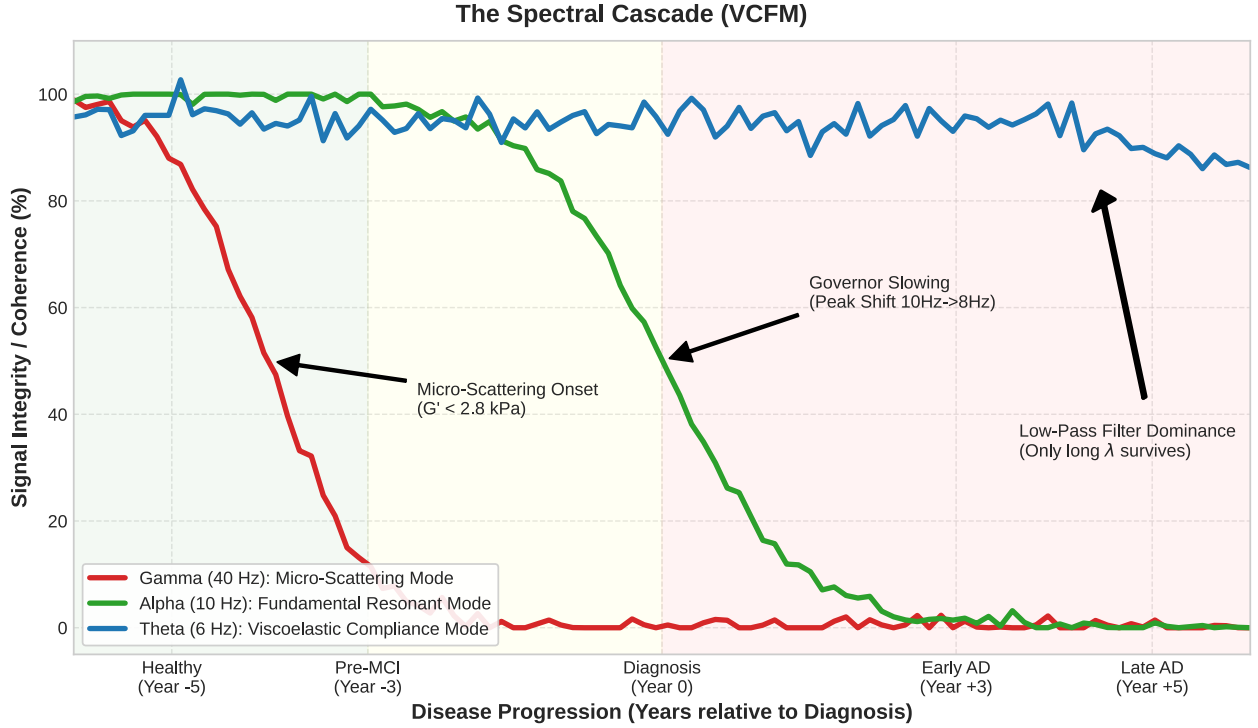


Figure 4: **Figure 4: The Spectral Cascade (VCFM).** Signal integrity over disease progression for three frequency modes. Gamma (40 Hz, red) shows earliest decline with Micro-Scattering Onset at $G' < 2.8$ kPa during Pre-MCI. Alpha (10 Hz, green) follows with Governor Slowing (peak shift 10Hz $\rightarrow$ 8Hz) around diagnosis. Theta (6 Hz, blue) persists as the low-pass filter allows only long wavelengths through degraded tissue. The temporal sequence (Year -5 to Year $+5$ relative to diagnosis) demonstrates the 3–5 year lead time of gamma loss over clinical presentation.

4.6 AD Biomarker Cascade

1. **10 years prior:** Amyloid accumulation (PET-positive, asymptomatic)
2. **5 years prior:** Tau spreading; GABAergic dysfunction; χ drops below 0.8
3. **2 years prior:** Hyperexcitability, epileptiform activity ($\sim 42\%$ of MCI)
4. **Diagnosis:** Memory impairment; compensatory inflammation
5. **2 years post:** Stiffness collapse ($G' < 2.2$ kPa); χ rises above 1.2

4.7 EEG Signature

Diffuse alpha slowing (f_α : 10 $\rightarrow$ 7 Hz) followed by theta/delta dominance. The spectral cascade follows:

$$\text{Gamma scattering} \rightarrow \text{Alpha detuning} \rightarrow \text{Theta dominance} \quad (9)$$

4.8 Unified Staging Thresholds

Table 2: Unified SymC staging thresholds for AD (approximate values derived from meta-analysis).

Stage	χ Range	G' (kPa)	Lead Time	Clinical
Healthy	0.85–0.95	> 3.0	N/A	Stable
Pre-MCI	0.70–0.80	2.8–3.0	3–5 yrs	Fluctuating
Early AD	1.10–1.30	< 2.5	0–2 yrs	Declining
Late AD	> 1.40	< 2.0	Past	Severe

Note the U-curve trajectory: Pre-MCI shows lowest χ (underdamped instability), while Late AD shows highest χ (overdamped rigidity).

5 Type 2 Failure: Lewy Body Dementia (The Damping Oscillator)

5.1 The DLB Paradox

DLB presents a critical test for the SymC framework. Patients exhibit:

- Profound cognitive fluctuations (80% to 90% prevalence)
- Visual hallucinations
- REM sleep behavior disorder (prodromal by 10 to 15 years)
- Yet: **Normal brain stiffness on MRE**

Key Evidence: ElSheikh et al. (2017) found that “patients with dementia with Lewy bodies did not show significant stiffness changes in any of the regions” on MRE, in contrast to AD and FTD which showed 6% to 11% cerebral softening [2]. This was confirmed by systematic review: “in dementia with Lewy bodies no significant differences in brain stiffness were found” [3].

5.2 SymC Interpretation: Stochastic Controller Failure

DLB is the “electrical” dementia. The structure (G') is sound, but the controller (γ) is governed by a stochastic process with pathologically high volatility. The damping ratio evolves according to:

$$d\chi = -\frac{1}{\tau}(\chi - \chi_{\text{set}})dt + \sigma dW_t \quad (10)$$

where dW_t is a Wiener process. In healthy aging, $\sigma/\tau \ll 1$ and the system remains near $\chi_{\text{set}} \approx 0.9$. In DLB, cholinergic and dopaminergic depletion causes $\sigma \uparrow$ and $\tau \uparrow$ (slower correction), yielding pathologically high variance:

$$\text{Var}[\chi_{\text{DLB}}] = \frac{\sigma^2 \tau}{2} \gg \text{Var}[\chi_{\text{healthy}}] \quad (11)$$

The $\chi(t)$ trajectory in DLB is not a monotonic drift but a **high-variance random walk** around a shifting mean:

$$\chi_{\text{DLB}}(t) = \bar{\chi}(t) + \eta(t), \quad \langle \eta^2 \rangle^{1/2} \gg \langle \eta^2 \rangle_{\text{AD}}^{1/2} \quad (12)$$

Clinically, this manifests as “good days and bad days” with rapid alternation (hours to days) rather than the gradual drift of AD. The structure supports recovery, but the controller cannot hold position.

5.3 DLB Biomarker Cascade

1. **15 years prior:** Autonomic prodrome (constipation, REM behavior disorder)
2. **10 years prior:** α -synuclein in gut/brainstem
3. **5 years prior:** EEG Dominant Frequency Variability (DFV) increases
4. **2 years prior:** Attentional lapses, daytime somnolence
5. **Diagnosis:** Visual hallucinations, Parkinsonian signs

5.4 Diagnostic Index: Dominant Frequency Variability

Unlike AD where alpha frequency slows monotonically, DLB alpha fluctuates moment to moment. The index is defined as:

$$\text{DFV} = \sqrt{\text{Var}[f_\alpha(t)]} \quad (13)$$

A DFV threshold distinguishes DLB from AD with high specificity even at prodromal stages [4]. Quantitative EEG shows theta/alpha ratios significantly higher in DLB than AD, correlating with hallucination severity [5].

5.5 EEG Signature

- Dominant rhythm in theta range (~ 5 to 7 Hz vs. AD's ~ 7 to 8 Hz)
- High temporal variability: alternating normal alpha and diffuse slowing
- Variability correlates with clinical fluctuation severity

6 Type 3 Failure: Vascular Dementia (The Impedance Staircase)

6.1 The Structural Cascade

VaD differs fundamentally from AD and DLB: it is not primarily an intrinsic neural feedback failure but an **extrinsic supply failure**. The substrate is vascular injury (infarcts, white matter lesions, microbleeds) rather than proteinopathy.

6.2 SymC Interpretation: The Reflection Coefficient

Vascular dementia represents a failure of impedance matching at the neurovascular interface. In transmission line physics, the energy transfer between two media is governed by the reflection coefficient:

$$R = \frac{Z_{\text{tissue}} - Z_{\text{vessel}}}{Z_{\text{tissue}} + Z_{\text{vessel}}} \quad (14)$$

where the acoustic impedance $Z = \sqrt{\rho G'}$ depends on density ρ and shear modulus G' .

In health, $Z_{\text{tissue}} \approx Z_{\text{vessel}}$ (matched load), yielding $R \approx 0$ and efficient energy transfer. In VaD:

- Vessel stiffening (arteriosclerosis): $G'_{\text{vessel}} \uparrow \Rightarrow Z_{\text{vessel}} \uparrow$
- Tissue softening (chronic hypoperfusion): $G'_{\text{tissue}} \downarrow \Rightarrow Z_{\text{tissue}} \downarrow$

As $|R| \rightarrow 1$, cardiac pulse waves undergo destructive interference at the vessel-tissue boundary, producing the **“Hammer Effect.”** The reflected energy causes: (1) vessel wall damage (microbleeds, arteriosclerosis), (2) transmitted shockwaves (lacunar infarcts in deep structures), and (3) pulsatile stress on white matter (leukoaraiosis).

The $\chi(t)$ trajectory follows a **step-function pattern**, with each vascular event adding permanent damping:

$$\chi_{\text{VaD}}(t) = \chi_0 + \sum_k \Delta\chi_k \cdot \Theta(t - t_k) \quad (15)$$

where Θ is the Heaviside step function and t_k marks each vascular event. Unlike AD's smooth U-curve, VaD is a *staircase* of irreversible steps.

6.3 VaD Biomarker Cascade

1. **10 years prior:** Vascular risk factor accumulation
2. **5 years prior:** White matter hyperintensities (Fazekas ≥ 1)
3. **2 years prior:** Gait slowing (< 0.8 m/s), urinary urgency
4. **Diagnosis:** Executive dysfunction, processing speed deficits
5. **Progression:** Stepwise decline with each infarct/event

6.4 Diagnostic Index: Fazekas-Executive Index

A composite index combines:

$$\text{FEI} = w_1 \cdot \text{Fazekas} + w_2 \cdot (1 - v_{\text{gait}}/v_0) + w_3 \cdot \Delta f_{\alpha}^{\text{focal}} \quad (16)$$

Key features:

- Fazekas score ≥ 2 indicates significant white matter burden
- Gait velocity < 0.8 m/s precedes cognitive diagnosis
- **Focal** (not diffuse) alpha slowing distinguishes from AD

6.5 Blood Pressure Variability

A unique VaD marker: blood pressure variability (BPV) correlates with cognitive decline independent of mean BP [8]. High BPV represents unstable “drive” to the cerebrovascular system:

$$\text{High BPV} \Rightarrow \text{Underdamped perfusion} \Rightarrow \text{Delirium episodes} \quad (17)$$

7 Type 4 Failure: Frontotemporal Dementia (The Topological Cliff)

7.1 The EEG Paradox

FTD presents another paradox: severe behavioral symptoms with *normal posterior EEG* for years. Standard EEG (which emphasizes occipital alpha) misses FTD because:

1. Pathology is focal to frontal/temporal regions
2. Posterior rhythm generators remain intact early
3. The “instability front” affects connectivity, not power

Studies confirm: FTD EEG “often normal in early FTD (no diffuse slowing)” while AD shows diffuse changes [6].

7.2 SymC Interpretation: Loss of Small-World Integration

FTD represents a **topological phase transition** in network organization. The healthy brain operates as a “Small World” network: high local clustering (segregation) combined with short path lengths (integration). FTD selectively destroys the long-range connections that enable integration while preserving local modules.

This can be quantified by the Kuramoto synchronization order parameter:

$$r(t) = \left| \frac{1}{N} \sum_{j=1}^N e^{i\theta_j(t)} \right| \quad (18)$$

where $\theta_j(t)$ is the phase of oscillator j . In health, frontal hubs maintain $r \approx 0.6$ to 0.8 (partial synchronization). In FTD:

$$r_{\text{frontal}}(t) \rightarrow 0 \quad (\text{decoherence}) \quad (19)$$

while posterior r remains near-normal.

The *global efficiency* of the network also degrades:

$$E_{\text{glob}} = \frac{1}{N(N-1)} \sum_{i \neq j} \frac{1}{d_{ij}} \rightarrow E_{\text{lattice}} \quad (20)$$

where d_{ij} is the shortest path length between nodes i and j . As frontal hubs disconnect, the network transitions from Small World (E_{glob} high) to fragmented lattice (E_{glob} low).

This explains the “Normal EEG / Abnormal Behavior” paradox: local oscillatory power (measured by standard EEG) is preserved because local modules function normally. But global integration, required for impulse control, social cognition, and behavioral regulation, collapses because the *topology* that connects modules is destroyed.

7.3 FTD Biomarker Cascade (Behavioral Variant)

1. **8 years prior:** Focal atrophy (amygdala, insula) on MRI
2. **5 years prior:** Empathy loss, eating behavior changes
3. **2 years prior:** Disinhibition, compulsive behaviors
4. **Diagnosis:** Frontal-temporal hypoconnectivity (not power loss)
5. **Late:** Global slowing emerges only in advanced stages

7.4 Diagnostic Index: Frontal Connectivity-Behavior Index

$$\text{FCBI} = w_1 \cdot \text{NPI}_{\text{disinhibit}} + w_2 \cdot (1 - C_{\text{F-T}}) + w_3 \cdot \mathbb{I}[\text{EEG normal}] \quad (21)$$

Key insight: A “normal” posterior EEG in a patient with dementia symptoms is a **positive predictor** of FTD over AD.

7.5 MRE Signature

Unlike DLB (normal stiffness), FTD shows regional softening in frontal and temporal lobes (6% to 7% decrease) [2]. This distinguishes FTD from DLB mechanically:

- **FTD:** Focal softening (frontal/temporal) + normal posterior EEG
- **DLB:** Normal stiffness + fluctuating EEG

8 Cognitive Reserve: Active Gain Control

8.1 The Reserve Problem

Why do some individuals with substantial pathological burden (amyloid, tau, vascular lesions) maintain cognitive function while others with less pathology decline rapidly? The SymC framework provides a mathematical definition of cognitive reserve.

8.2 SymC Interpretation: Metabolic Compensation

Cognitive reserve is defined as **Active Gain Control**: the capacity to increase metabolic drive (ω) to compensate for accumulating damping (γ):

$$\chi_{\text{reserve}}(t) = \frac{\gamma(t) + \Delta\gamma_{\text{pathology}}}{2(\omega_0 + \Delta\omega_{\text{compensation}})} \quad (22)$$

A high-reserve brain maintains $\chi \approx 0.9$ by increasing ω as γ rises. Clinical stability persists until the **metabolic ceiling** is reached:

$$\omega_{\text{max}} = \omega_0 + \Delta\omega_{\text{ceiling}} \quad (23)$$

When γ exceeds what ω_{max} can compensate, χ rapidly deviates and clinical symptoms emerge. This explains the phenomenon of sudden decline after prolonged stability.

8.3 Reserve Depletion Dynamics

The reserve depletion rate depends on pathological load:

$$\frac{d\omega_{\text{available}}}{dt} = -k \cdot \gamma_{\text{pathology}}(t) \quad (24)$$

High-reserve individuals have larger ω_{ceiling} and slower depletion rate k , extending the preclinical window.

9 Mixed Dementia: Composite Failure

9.1 The Synergy Problem

Mixed dementia (typically AD + VaD) represents simultaneous engagement of multiple failure modes. The clinical trajectory is often:

- More rapid decline than either pathology alone
- Lower threshold for decompensation
- Narrower adaptive window

Studies confirm: “each pathology contributed independent variance to cognitive impairment” with roughly additive effects [7].

9.2 SymC Interpretation: Multi-Mode Failure

In mixed dementia, multiple χ_i deviate simultaneously:

$$\chi_{\text{mixed}} = f(\chi_{\text{AD}}, \chi_{\text{VaD}}, \dots) \quad (25)$$

The effective adaptive window narrows:

$$\Delta\chi_{\text{safe}}^{\text{mixed}} < \min(\Delta\chi_{\text{safe}}^{\text{AD}}, \Delta\chi_{\text{safe}}^{\text{VaD}}) \quad (26)$$

9.3 Diagnostic Approach

Mixed dementia requires multi-modal assessment:

- Amyloid/tau biomarkers (AD component)
- White matter burden (VaD component)
- EEG: more pronounced slowing than pure AD or VaD alone

10 Synthesis: The Universal Instability Map

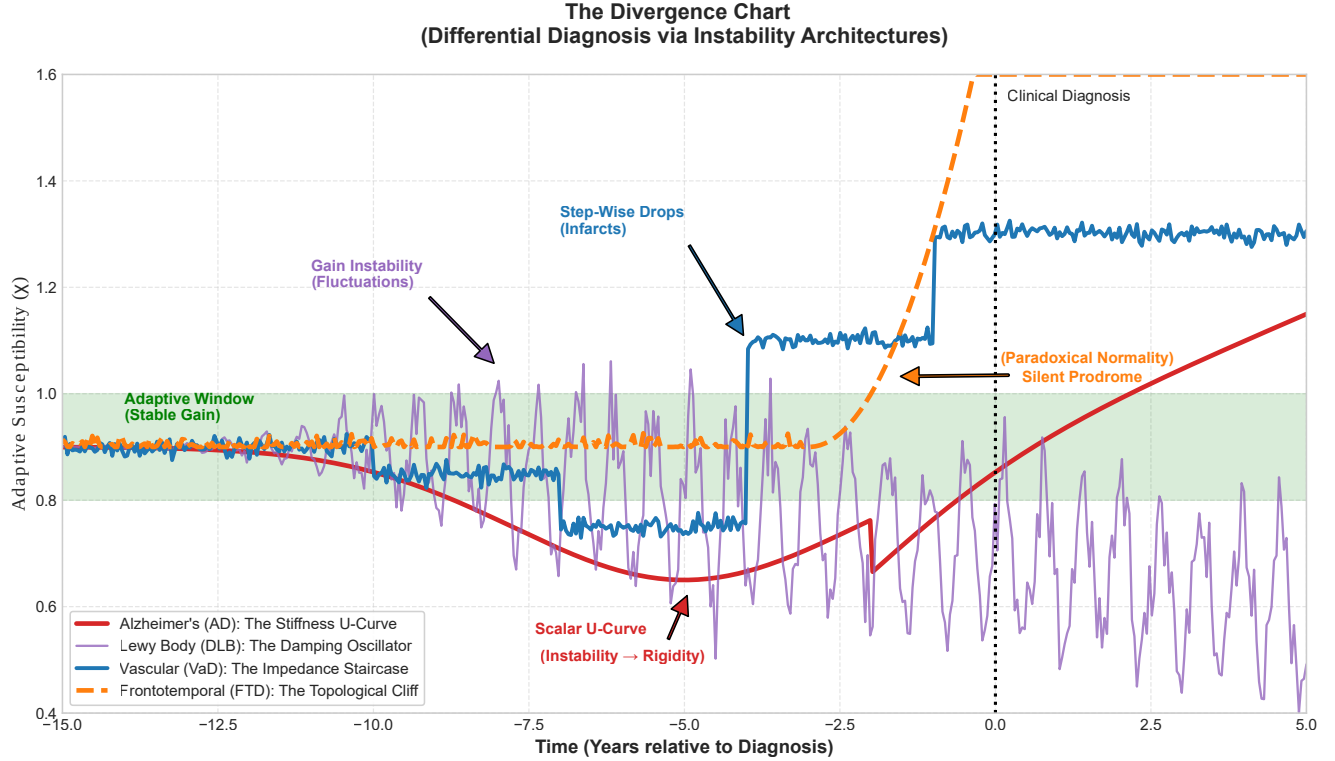


Figure 5: **Figure 5: The Divergence Chart.** Each dementia subtype follows a distinct $\chi(t)$ trajectory relative to clinical diagnosis (Year 0). AD (red) exhibits a U-curve from underdamped instability to over-damped rigidity. DLB (purple) shows high-variance oscillations around the adaptive window. VaD (blue) follows a step-function pattern driven by discrete vascular events. FTD (orange) maintains paradoxical normality until sudden topological collapse. The green band marks the adaptive window ($\chi \approx 0.8$ to 1.0).

Table 3: SymC Mapping of Dementia Subtypes

Feature	AD	DLB	VaD	FTD
Failure Type	Type 1	Type 2	Type 3	Type 4
Primary Failure	Stiffness (G')	Damping (γ)	Impedance (ΔZ)	Topology ($\mathcal{C}$)
$\chi(t)$ Pattern	U-curve	High variance	Step-function	Domain-split
EEG Signature	Diffuse slowing	High DFV	Focal slowing	Normal (early)
MRE Finding	Global softening	Normal	Variable	Focal softening
Lead Marker	Memory	RBD/Autonomic	Gait/Urinary	Behavior
Preclinical Window	3–5 yrs	10–15 yrs	Variable	5–8 yrs

10.1 Divergence Points

The key diagnostic divergence points are:

1. **MRE:** If stiffness is *normal* $\Rightarrow$ not AD/FTD $\Rightarrow$ consider DLB
2. **EEG Power:** If posterior alpha is *normal* $\Rightarrow$ not AD $\Rightarrow$ consider FTD
3. **EEG Variability:** If DFV is *high* $\Rightarrow$ DLB over AD
4. **Stepwise Course:** If decline follows vascular events $\Rightarrow$ VaD component

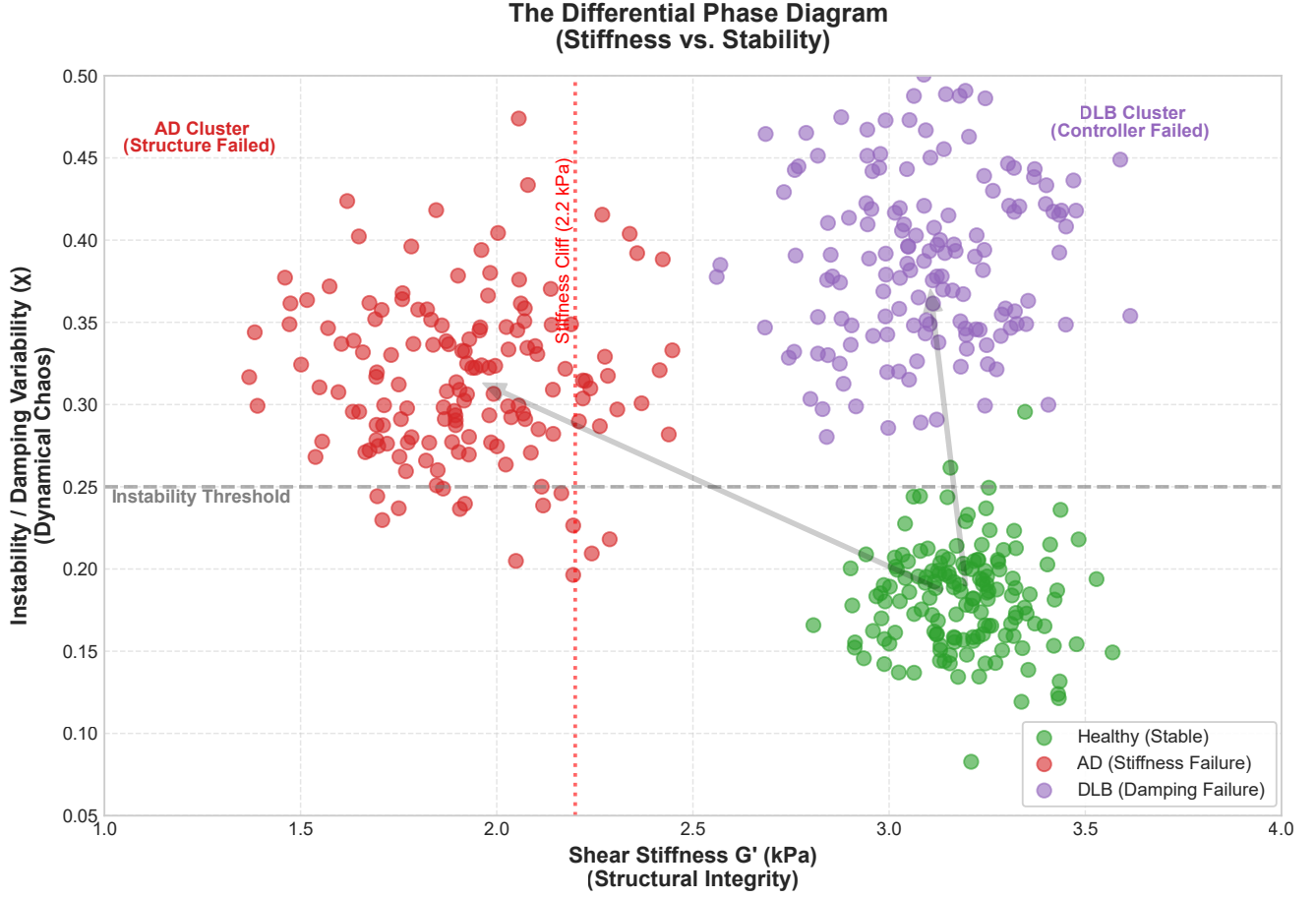


Figure 6: **Figure 6: The Differential Phase Diagram.** Scatter plot of shear stiffness (G') versus damping variability (χ) for healthy controls (green), AD (red), and DLB (purple). AD clusters in the low-stiffness/moderate-variability region (structure failed). DLB clusters in the normal-stiffness/high-variability region (controller failed). Healthy controls occupy the high-stiffness/low-variability quadrant. The vertical dashed line marks the stiffness cliff ($G' = 2.2$ kPa); the horizontal dashed line marks the instability threshold. This two-dimensional representation enables single-visit differential diagnosis.

11 Multi-Mode χ Aggregation

11.1 The Aggregation Problem

A single patient may have $\chi_{\text{motor}} = 1.1$ (slightly overdamped) and $\chi_{\text{cognitive}} = 0.7$ (underdamped). How do we represent overall status?

11.2 Proposed Approaches

Option 1: Weakest-Link Model

$$\chi_{\text{eff}} = \chi_i \text{ where } |\chi_i - 1| = \max_j |\chi_j - 1| \quad (27)$$

Clinical status determined by worst-affected domain.

Option 2: Weighted Composite

$$\chi_{\text{composite}} = \sum_i w_i \chi_i, \quad \sum_i w_i = 1 \quad (28)$$

Weights reflect functional importance of each domain.

Option 3: Vector Representation

$$\vec{\chi}(t) = (\chi_{\text{memory}}, \chi_{\text{exec}}, \chi_{\text{motor}}, \chi_{\text{sleep}}, \chi_{\text{autonomic}}) \quad (29)$$

Track each domain independently; failure when any component crosses threshold.

The five χ -components capture distinct functional domains:

- χ_{memory} : Hippocampal-dependent encoding/retrieval (AD-dominant)
- χ_{exec} : Frontal executive control (FTD/VaD-dominant)
- χ_{motor} : Basal ganglia motor regulation
- χ_{sleep} : Brainstem sleep architecture (DLB-prodromal)
- $\chi_{\text{autonomic}}$: Peripheral/central autonomic function (DLB-prodromal)

For clinical practice, **Option 3 with threshold alerts** is recommended: monitor the full vector and intervene when any $|\chi_i - 1| > 0.3$.

12 Predictive Windows: The 3–15 Year Detection Horizon

The SymC framework enables subtype-specific prediction models based on the biomarker cascade timing:

12.1 Type 1 (AD): 3–8 Year Window

The AD predictive cascade:

1. **Year –8**: Amyloid accumulation (PET-detectable, χ normal)
2. **Year –5**: ξ rise (damping ratio increases before stiffness drops)
3. **Year –3**: Gamma scattering onset ($G' \approx 2.8$ kPa)
4. **Year –1**: Alpha detuning (10 Hz $\rightarrow$ 8 Hz)
5. **Year 0**: Clinical diagnosis

12.2 Type 2 (DLB): 10–15 Year Window

The DLB predictive cascade:

1. **Year –15**: REM sleep behavior disorder onset
2. **Year –10**: Autonomic dysfunction (constipation, orthostasis)
3. **Year –5**: DFV > 1.2 Hz threshold crossed
4. **Year –2**: Cognitive fluctuations emerge
5. **Year 0**: Clinical diagnosis (hallucinations, parkinsonism)

12.3 Type 3 (VaD): 5–12 Year Window

The VaD predictive cascade:

1. **Year –12**: Vascular risk factor accumulation (BP variability)
2. **Year –8**: White matter hyperintensities (Fazekas 1)
3. **Year –5**: Gait slowing (< 0.8 m/s)
4. **Year –2**: First stepwise χ drop (minor infarct)
5. **Year 0**: Clinical diagnosis (executive dysfunction)

12.4 Type 4 (FTD): 5–10 Year Window

The FTD predictive cascade:

1. **Year −10:** Focal atrophy (amygdala, insula)
2. **Year −5:** Empathy loss, eating behavior changes
3. **Year −3:** Frontal-temporal coherence decline ($r_{\text{frontal}} \downarrow$)
4. **Year −1:** Disinhibition, compulsive behaviors
5. **Year 0:** Clinical diagnosis (normal posterior EEG)

13 Therapeutic Implications

13.1 Phase-Matched Intervention

The failure type determines the therapeutic target:

- **Type 1 (AD):** Restore Structure (G') and Drive (ω): anti-amyloid, anti-inflammatory, gamma entrainment
- **Type 2 (DLB):** Stabilize Gain (γ): cholinesterase inhibitors, avoid dopamine blockers
- **Type 3 (VaD):** Match Impedance (lower Z_{vessel}): vascular risk control, BP variability reduction
- **Type 4 (FTD):** Maintain Connectivity ($\mathcal{C}$): SSRIs for disinhibition, stimulants for apathy

The key insight: **phase-matched intervention**. Treating an overdamped system with further damping (sedatives) or an underdamped system with further excitation (stimulants) worsens outcomes. Therapy must push χ toward the adaptive window from the correct direction.

13.2 Therapy-Agnostic Staging

The SymC framework applies an oncology-style staging logic to neurology. Just as oncology distinguishes between proliferative (high-growth) and necrotic (dormant) tumor stages to determine treatment, the SymC framework distinguishes between high-energy instability (underdamped) and low-energy rigidity (overdamped).

Current AD trials fail because they treat the disease as a monolith. Applying “damping” agents (anti-excitatory drugs) to a brain that is already “rigid” (Overdamped Phase 2) is mechanically destined to fail, analogous to treating a necrotic tumor with anti-proliferatives. Success is defined by boundary physics: does the intervention steer $\chi(t)$ back to the adaptive window appropriate for the patient’s specific dynamical stage?

13.3 Closed-Loop Adaptive Control

This architecture mirrors the success of Adaptive Deep Brain Stimulation (aDBS) in Parkinson’s disease [10]. Just as aDBS monitors beta-band power to titrate stimulation, the SymC framework titrates intervention to maintain χ within the adaptive window. The goal is not just pathology removal but active maintenance of the brain’s dynamical operating point.

14 Falsifiable Predictions

14.1 AD-Specific Predictions

1. **The Lead-Time Principle:** Increases in ξ will be detectable before drops in G' or volumetric atrophy. Estimated lead time: 3 to 5 years.
2. **The VCFM Spectral Cascade:** Failure will follow the sequence: Gamma scattering ($G' \approx 2.8$ kPa) $\rightarrow$ Alpha detuning ($G' \approx 2.5$ kPa) $\rightarrow$ Theta dominance ($G' < 2.0$ kPa).

3. **U-Curve Trajectory:** Longitudinal monitoring will show converters traversing: Healthy ($\chi \approx 0.9$) $\rightarrow$ Underdamped ($\chi < 0.8$) $\rightarrow$ Overdamped ($\chi > 1.2$).
4. **Cross-Domain Concordance:** MRE-derived ξ and EEG-derived ζ will converge when expressed as χ ($r > 0.7$).

14.2 Cross-Diagnostic Predictions

1. **DLB MRE Prediction:** Longitudinal MRE in prodromal DLB (RBD+ cohorts) will show preserved stiffness even as clinical symptoms emerge, in contrast to AD.
2. **FTD EEG Prediction:** Frontal-temporal coherence (not power) will decline 3 to 5 years before posterior alpha slowing in bvFTD.
3. **VaD BPV Prediction:** Reducing blood pressure variability (not just mean BP) will slow cognitive decline in subcortical VaD.
4. **Composite Index:** A composite index combining MRE stiffness, EEG variability, and connectivity will outperform single-modality biomarkers for differential diagnosis.
5. **Therapeutic Mediation:** Clinical improvement will be statistically mediated by χ normalization.

14.3 Falsification Criteria

The framework is falsified if any of the following are observed:

- Patients convert to AD without any preceding χ deviation from the adaptive window
- The U-curve trajectory is absent (χ monotonically increases or decreases throughout disease)
- The spectral cascade does not follow the predicted sequence (gamma $\rightarrow$ alpha $\rightarrow$ theta)
- MRE-derived ξ and EEG-derived ζ show no concordance when expressed as χ
- ξ does not precede G' collapse (i.e., stiffness falls before damping rises)
- 40Hz entrainment efficacy shows no correlation with tissue stiffness
- Clinical improvement occurs without normalization of SymC indices
- DLB patients show stiffness collapse equivalent to AD
- FTD patients show diffuse EEG slowing before behavioral symptoms

15 Conclusion

The Symmetrical Convergence framework reveals that dementia is not a unitary phenomenon. Each subtype represents a distinct “Instability Architecture,” a specific pattern of failure in the variables governing dynamical stability:

- **Type 1 (AD):** Stiffness collapse $\Rightarrow$ U-curve $\chi(t)$
- **Type 2 (DLB):** Controller failure $\Rightarrow$ High-variance $\chi(t)$
- **Type 3 (VaD):** Impedance mismatch $\Rightarrow$ Step-function $\chi(t)$
- **Type 4 (FTD):** Topology failure $\Rightarrow$ Domain-split $\chi(t)$

We must stop classifying dementia solely by the proteins found at autopsy (the debris) and start classifying by the dynamics observed in life (the failure). By introducing χ as a universal translation layer, the framework reveals that the damping ratio in control theory (ζ), the viscoelastic loss factor in materials science (ξ), and the stability fluctuations observed clinically are all measuring the same quantity. The central equation $\xi = G''/(2G') \equiv \chi$ unifies tissue mechanics and neural dynamics under a single governing principle.

The Viscoelastic Cascading Failure Model (VCFM) provides the mechanism for Type 1: neural tissue acts as a low-pass filter, with stiffness decline producing a predictable spectral cascade that explains why EEG slowing is among the earliest predictors of AD conversion. For non-AD dementias, distinct failure modes produce distinct trajectories, enabling physics-based differential diagnosis years before clinical phenotypes converge.

The predictions are falsifiable and the measurement tools exist. If validated, the SymC framework establishes a physics-based roadmap for understanding neurodegeneration not as “loss of brain” but as “loss of stability”: a problem amenable to detection, measurement, and potentially, correction.

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